

Linear Convolution	Circular Convolution
$y[n] = h[n] \otimes x[n]$	$y[n] = h[n] \otimes_N x[n]$
$y[n] = \sum_{m=0}^{N-1} h[m]x[(n - m)]$	$y[n] = \sum_{m=0}^{N-1} h[m]x[((n - m))_N]$
$x[n] = [1,2,3], h[n] = [1,2,3,4,5]$ $y[n] = [\textcolor{teal}{1}, \textcolor{red}{4}, 10, 16, 22, \textcolor{teal}{22}, \textcolor{red}{15}]$	$x[n] = [1,2,3,0,0], h[n] = [1,2,3,4,5]$ $y[n] = [23,19,10,16,22]$